# Design Brief – Review 2

## 0. Header

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|---|---|
| **Project code** | FA26SE032 |
| **Project name** | Sales Collaborator and Affiliate Network Management System (SCANMS) |
| **Team** | Nguyễn Thành Thắng (Leader - SE184251), Nguyễn Đình Tuấn (SE182540), Nguyễn Phú Quý (SE180104), Phan Xuân Thịnh (SE184527) |
| **Supervisor** | ThS. Tôn Thất Hoàng Minh |
| **Repository link** | https://github.com/thang-nt25/SCANMS-project-capstone-fall26 (tag: review-2) |
| **Version** | 2.0 |
| **Date** | 06/11/2026 |

## 1. Scope

### 1.1 Registration Changes

| Section | Current | Proposed | Reason |
|---|---|---|---|
| **Timeline & Team (Sec. 1–2)** | 09/2026–03/2027; 5th unassigned row | 07/09/2026–30/12/2026; finalized 4 members | Match Fall 2026 defense calendar and lock team roster. |
| **Domain Model & Pricing (Sec. 3.2a–b)** | Open multi-seller portal; unguarded discounts | 4 in-house brand divisions, Category Lock, EAN-13 check & MAP floor-price promotions | Focus strictly on enterprise collaborator management, not an open C2C marketplace. |
| **5-Role IAM & SoD (Sec. 3.2b–c)** | 3 roles; public shop signup; guest checkout | 5 roles; Admin-only staff provisioning; KOC eKYC (Bronze 10%); 0-loss auth checkout | Prevent external shop creation and ensure verified tax/delivery identity. |
| **Remuneration & Attribution (Sec. 3.2b)** | Simple/additive rates; generic vouchers | Best-Benefit Precedence (cap 25%), KOC coupon [KOC][5\|10], Self-Referral = 0%, KPI escrow | Protect brand margins, ensure cookie-less attribution, and block self-buying. |
| **Returns, Escrow & Payout (Sec. 3.2b–c)** | Linear payout; no return split; manual file only | 14-day hold; 1-1 Exchange (keeps KOC fee) vs Refund (clawback); 10% PIT & BIDV Sandbox | Fairly protect KOCs on carrier damage while preventing overpayment on refunds. |
| **Realtime, AI, Cloud & Mobile (Sec. 3.2b,f)** | Static web portal; no chat/SLA/AI/mobile | Socket.IO VIP deals (>16% Admin gate), 14d Sample SLA, Dual AI, Cloud Web + Expo Mobile App | Support real-time partner workflows across deployed Web and Android clients. |

### 1.2 Out of Scope

| Item | Reason |
|---|---|
| **Open External Seller Marketplace & Public Shop Registration** | SCANMS governs external collaborators for in-house enterprise brand divisions; external third-party merchants cannot open public stores. |
| **Generic Retail Voucher Stacking & Standalone Marketing Portals** | Platform/freeship voucher stacking and standalone promotional portals are excluded to prevent margin cannibalization; only MAP-guarded ProductPricePromotion and KOC-bound attribution coupons are supported. |
| **Factory Manufacturing, Supplier Procurement (ERP/AP) & Robotics WMS** | Upstream OEM manufacturing, purchase orders, and warehouse hardware belong to corporate ERP; SCANMS manages sellable stock quotas and SKU floor prices. |
| **Proprietary Courier Vehicles & Licensed Core Banking Rails** | Physical parcel delivery and fiat banking rails are delegated to GHN Sandbox API, PayOS (order collection), and BIDV Sandbox API (collaborator payouts). |

## 2. System at a Glance

SCANMS solves fragmented partner onboarding, attribution fraud, margin erosion, and manual commission settlement for multi-brand D2C retail enterprises managing external sales collaborators (KOLs/KOCs). Built for five roles—System Administrator, System Manager, In-House Shop Manager, Collaborator, and Customer—across deployed Web portals (React/Vite) and a native Android application (React Native/Expo), SCANMS automates the entire affiliate lifecycle across four company-owned brand divisions. Core capabilities include Admin-governed staff provisioning, Maker-Checker eKYC, category-locked catalogs, MAP floor-price promotions, 4-axis AI partner matchmaking, Socket.IO VIP deal negotiation with >16% Admin budget escalation, 14-day sample review SLAs, Smart Link/QR and KOC-coupon attribution, 14-day post-delivery Adaptive Escrow ledger matching customer return warranty with 1-for-1 exchange vs. refund clawback, and row-locked payouts with 10% PIT withholding via BIDV Sandbox. See System Overview Architecture Diagram in SDD (Report 4, Section 3.1).

## 3. Key Design Decisions

| ID | Decision | Options considered | Chosen option and rationale | Owner | Related FR / diagram |
|---|---|---|---|---|---|
| **D1** | In-House Brand Provisioning & Shared Mobile/Web Bootstrap | Public seller self-signup & separate mobile BFF vs Admin-only staff provisioning & unified NestJS API | Blocked public SHOP_MANAGER signup (/admin/personnel only) and exposed /api/mobile/bootstrap over one NestJS modular monolith so Web and Expo Android enforce identical 5-role RBAC. | Nguyễn Thành Thắng | F-01, F-02 / SDD Sec. 3.1 |
| **D2** | MAP Floor-Price Guard & KOC-Exclusive Attribution Coupon | Unrestricted shop price edits & multi-voucher stacking vs Maker-Checker MAP guard & KOC-bound coupon | Enforced proposedPrice >= floorPrice with System Manager approval (ProductPricePromotion) and single KOC goods coupon ([KOC][5\|10]) for deterministic cookie-less attribution. | Nguyễn Thành Thắng | F-06, F-13 / SDD Sec. 4.2 |
| **D3** | Socket.IO VIP Deal Negotiation with >16% Admin Escalation | HTTP polling & unguarded deal rates vs WebSocket rich deal cards with 2-tier approval gate | Used Socket.IO /chat rooms with autonomous activation for 10%–16% deals and mandatory PENDING_ADMIN_APPROVAL escalation to System Admin for 17%–25% deals. | Phan Xuân Thịnh | F-09, F-10 / SDD Sec. 5.3 |
| **D4** | 14-Day Post-Delivery Hold & Dual-Path Return Resolution | Instant commission release or blanket clawback on all claims vs 14-day hold with Exchange vs Refund split | Held pending commissions 14 days post-delivery: 1-for-1 EXCHANGE on carrier damage preserves 100% KOC commission (+ GHN claim), whereas REFUND triggers automatic clawback. | Nguyễn Phú Quý | F-16, F-17 / SDD Sec. 5.4 |
| **D5** | Best-Benefit Commission Precedence & Row-Locked BIDV Payout | Additive rate stacking & unlocked balance updates vs Precedence override, KPI escrow & SELECT FOR UPDATE | Applied min(25%, max(VIP Deal, 10%+Tier, 10%) + Live), isolated VIP bonus in conditionalDealEscrowBalance, and locked wallet rows with 10% flat PIT withholding via BIDV Sandbox. | Nguyễn Đình Tuấn | F-17, F-19 / SDD Sec. 6.2 |

## 4. Top Risks

| ID | Risk | Impact | How we verify or mitigate | Owner |
|---|---|---|---|---|
| **R1** | Self-Referral Purchases & Bot Click Flooding distorting attribution and draining marketing budgets. | High: Unfair commission leakage and skewed AI conversion rates (CR%). | isSelfReferral guard forces self-purchase commission to 0%; Redis sliding-window limit (100 req/min/IP) with RAM LRU fallback and AiFraudService anomaly flags. | Nguyễn Phú Quý |
| **R2** | Brand Margin Erosion from below-cost discounts or excessive VIP collaborator rates. | High: Negative unit economics on discounted SKUs. | assertFloor(proposedPrice, floorPrice) Maker-Checker approval, Best-Benefit non-additive cap (<=25%), >16% Admin approval gate, and KPI-locked deal bonus escrow. | Nguyễn Thành Thắng |
| **R3** | Concurrent Withdrawal Double-Spending, Unreturned Samples, or Refund Overpayment. | High: Wallet overdrafts, sample inventory loss, and unrecovered commissions. | PostgreSQL SELECT ... FOR UPDATE wallet locks, idempotent BIDV Sandbox inquiry, 14-day return hold clawback, and 14-day sample SLA overdue quota lock. | Nguyễn Đình Tuấn |

## 5. Known Gaps

| ID | Gap | Location | Plan | Owner | Due |
|---|---|---|---|---|---|
| **G1** | SDD ERD & Architecture Docs (21-table draft) lag behind 69-model Prisma Schema & Expo Mobile client (BM02 SR-01, SR-02) | docs/architecture/SAD_DOCUMENT.md, DATABASE_DESIGN.md vs schema.prisma & Mobile/ | Regenerate Logical ERD, State Machines, and System Overview diagram in Report 4 to match 69 Prisma models, Vite Web, Expo Mobile, and @nestjs/schedule. | Nguyễn Thành Thắng | Defense 1.1 |
| **G2** | Mobile Android App defers Customer PayOS Checkout & Live Room to Web Portal in current iteration (BM02 SR-04) | Mobile/src/features/cart/CartScreen.tsx, Mobile/docs/ROLE_WORKFLOWS_2026-11-03.md | 5-role Mobile operations (catalog, KYC, chat, VIP deals, samples, pricing, GHN orders, returns, payouts) are active; complete native checkout deep-link handoff & staging APK build. | Phan Xuân Thịnh | Defense 1.1 |
| **G3** | Commercial Host-to-Host Bank Disbursement simulated via BIDV Sandbox Adapter (BM02 SR-05) | backend/src/modules/payouts/bidv-payout.adapter.ts | Kept by design due to banking license constraints; BIDV Sandbox idempotency, inquiry reconciliation (SBX-...), FinancePosting, and VietQR batch export verified. | Nguyễn Đình Tuấn | Kept (Sandbox) |
| **G4** | Residual deprecated database/mobile stubs after scope removal of standalone promotional portals (BM02 SR-03) | backend/prisma/schema.prisma, Mobile/src/features/workspace/policy.ts | Web routes already redirect/exclude legacy flows and rate parameter is fixed to 0; hide remaining mobile menu entry and mark legacy DB table deprecated in SDD. | Nguyễn Thành Thắng | Defense 1.1 |

## 6. Proposed Mandatory Features

| ID | Feature | Actor | Acceptance criteria (testable) | Test | Status |
|---|---|---|---|---|---|
| **F-01** | Multi-Role Auth, Registration/Reset Email OTP & 5-Role RBAC (Web & Mobile) | All 5 Roles | Argon2id (>=64MB) hashes passwords; >=5 wrong OTPs invalidate code; unauthorized Web routes or /api/mobile/bootstrap calls return 403 Forbidden. | TC-F01-01..03 | **Done** |
| **F-02** | Internal Brand Staff & In-House Store Provisioning | System Administrator | Only SYSTEM_ADMIN creates SHOP_MANAGER (bound to storeId & GHN warehouse) and SYSTEM_MANAGER; public /auth/register with SHOP_MANAGER coerces to CUSTOMER. | TC-F02-01..03 | **Done** |
| **F-03** | KOC Upgrade eKYC Maker-Checker & Social Channels | Customer, Collaborator, System Manager | Customer submits 12-digit CCCD, bank account & TikTok/FB/YouTube URL; SYSTEM_MANAGER approval sets KycStatus=VERIFIED, role=COLLABORATOR, starting tier=Bronze (10%). | TC-F03-01..03 | **Done** |
| **F-04** | Immutable Security & Financial Audit Trail | System Administrator, System Manager | 100% of staff creations, KYC decisions, MAP updates, VIP deal approvals, return resolutions & payouts append an AuditLog entry; API blocks edit/delete. | TC-F04-01..03 | **Done** |
| **F-05** | Category-Locked 6-Block Product & Compliance Moderation | Shop Manager, System Manager | Product category locks to store.category (mismatch returns 400); duplicate EAN-13 barcode fails; new SKU stays DRAFT until a System Manager approves (APPROVED). | TC-F05-01..03 | **Done** |
| **F-06** | MAP Floor-Price Guard & Price Promotion Approval | Shop Manager, System Manager, System Administrator | System Manager sets SKU floorPrice; single/batch promotion with proposedPrice < floorPrice returns 400; discount applies only when status=ACTIVE within budget. | TC-F06-01..03 | **Done** |
| **F-07** | 4-Axis AI Partner Matchmaking & Two-Way Cooperation | Shop Manager, Collaborator | AI scores verified KOCs (0–100 across Category, CR%, Tier/Social, Price Fit) in <=500ms; bilateral partnership invite/request transitions PENDING -> APPROVED. | TC-F07-01..03 | **Done** |
| **F-08** | Centralized Digital Media Asset Hub | Shop Manager, Collaborator | Shop Manager uploads product-linked HD images/videos/copy; verified Collaborator filters by type and downloads in 1 click; unverified access returns 403. | TC-F08-01..03 | **Done** |
| **F-09** | Socket.IO Realtime VIP Deal Negotiation (10%–16%) | Shop Manager, Collaborator | Counterparties send/respond to rich VIP Deal cards (targetRevenue, commitmentDays) over /chat WebSocket without reload; 10%–16% deals activate immediately on accept. | TC-F09-01..03 | **Done** |
| **F-10** | VIP Budget Exception Escalation Gate (17%–25%) | Shop Manager, Collaborator, System Administrator | Proposals >25% fail with 400; accepted 17%–25% deals transition to PENDING_ADMIN_APPROVAL and activate only after SYSTEM_ADMIN approves at /admin/vip-exceptions. | TC-F10-01..03 | **Done** |
| **F-11** | Sample Product Sponsorship & 14-Day Video Review SLA | Collaborator, Shop Manager, System Manager | Verified KOC requests 0 VND sample; GHN receipt (RECEIVED) starts 14-day SLA timer; unsubmitted video past deadline marks OVERDUE and blocks new sample requests. | TC-F11-01..03 | **Done** |
| **F-12** | Value-Added Livestream Room & Live Deal Pinning | Collaborator, Shop Manager, Customer | Creator transitions SCHEDULED -> LIVE -> ENDED and pins store SKUs (+2% live bonus); stock=0 disables buy CTA in live tray; ENDED locks pinning and shows summary. | TC-F12-01..03 | **Done** |
| **F-13** | Smart Link, Dynamic QR & KOC Attribution Coupon | Collaborator, Shop Manager, Customer | Verified KOC creates /r/:shortCode link, QR & goods coupon ([KOC][5\|10], shipping vouchers excluded); click sets 30d AttributionSession rate-limited at 100 req/min/IP. | TC-F13-01..03 | **Done** |
| **F-14** | Authenticated Zero-Loss Checkout & Self-Referral Guard | Customer, Collaborator | Checkout requires login while preserving 100% cart items and settles via PayOS VietQR/Wallet/COD; if buyer matches KOC (isSelfReferral), commission is forced to 0%. | TC-F14-01..03 | **Done** |
| **F-15** | Multi-Warehouse GHN Fulfillment & Order Tracking | Shop Manager, Customer | Shop Manager confirms order and dispatches GHN parcel from brand warehouse (weight/dimensions); SKU stock decrements atomically and status tracks SHIPPING -> DELIVERED. | TC-F15-01..03 | **Done** |
| **F-16** | Dual-Path Return Resolution: Exchange vs Refund Clawback | Customer, Shop Manager, System Manager | Within 14-day return window & 48h inspection SLA: EXCHANGE ships 1-1 replacement, preserves 100% KOC commission & opens GHN claim; REFUND credits wallet & reverses commission. | TC-F16-01..03 | **Done** |
| **F-17** | Best-Benefit Commission, 14-Day Hold & KPI Deal Escrow | Shop Manager, Collaborator, System Manager | Computes min(25%, max(VIP Deal, 10%+Tier, 10%) + Live); holds base fee 14 days in pendingBalance and holds VIP bonus in conditionalDealEscrowBalance until KPI QUALIFIED. | TC-F17-01..03 | **Done** |
| **F-18** | Monthly Milestone Settlement (FR-09) & Tier Promotion | Shop Manager, Collaborator | Closed-month clean GMV scan excludes refunded/self-referral orders, pays 200k–4M VND milestone bonus idempotently (0 duplicates), and promotes Bronze->Silver->Gold->Diamond. | TC-F18-01..03 | **Done** |
| **F-19** | Row-Locked Collaborator Withdrawal, 10% PIT & BIDV Payout | Collaborator, System Manager, System Administrator | SHOP_MANAGER withdrawal returns 403; COLLABORATOR request (>=200,000 VND) locks wallet row, withholds flat 10% PIT (FinancePosting), and disburses net via BIDV Sandbox. | TC-F19-01..03 | **Done** |

## 7. Non-Functional Requirements

| ID | Quality attribute | Metric | Threshold | Measurement method |
|---|---|---|---|---|
| **NFR-01** | API & Redirect Latency | REST & Smart Link p95 Response Time | <= 200 ms for catalog queries and /r/:shortCode redirection | Autocannon benchmark (500 concurrent requests) & NestJS timing interceptor |
| **NFR-02** | Authentication & RBAC Security | Password Hash Cost & Role Isolation | Argon2id (>= 64 MB memory, 3 iterations); 100% cross-role calls blocked (403) | Jest security suites (auth-otp.spec.ts, rbac guards) across Web & Mobile endpoints |
| **NFR-03** | Web & Mobile Client Responsiveness | Web Initial Bundle / FCP & Mobile Bootstrap | Web gzipped entry <= 500 KB, FCP <= 1.5 s; Android /mobile/bootstrap <= 300 ms | Vite build bundle audit, Lighthouse CI, and Mobile contract/startup test suite |
| **NFR-04** | Transactional & Inventory Integrity | Atomic Checkout & Return Rollback Rate | 100% atomic rollback on stock exhaustion or invalid return state transition | Jest integration specs (orders.service.spec.ts, return-pickup.postgres.integration.spec.ts) |
| **NFR-05** | Rate Limiting & Cache Resilience | Click Throttle & Redis Outage Failover | 100 req/min/IP limit; 0 unhandled 500 errors during simulated Redis disconnect | CacheService LRU memory fallback unit tests & click-flood load script |
| **NFR-06** | Financial Concurrency & Auditability | Double-Spend Prevention & Ledger Coverage | 0 negative wallet balances under 20 parallel payouts; 100% mutations in AuditLog | Parallel SELECT ... FOR UPDATE payout concurrency test & FinancePosting assertions |

## 8. Ownership Map

| Member | Role | Owned parts (2–3) | Main workflow owned | Key decisions / PRs |
|---|---|---|---|---|
| **Nguyễn Thành Thắng (Leader)** | Fullstack Architect & Platform Governance Lead | 5-Role IAM & Staff Provisioning (F-01..F-04), Two-Stage Escrow Ledger & 10% PIT Payouts (F-17..F-19) | WF-01: Internal Brand Staff Provisioning & KOC eKYC; WF-05: Two-Stage Escrow Maturation & 10% PIT Payouts | Decisions D1 & D5; System Governance, IAM, Ledger & Executive Dashboard |
| **Phan Xuân Thịnh** | Fullstack Developer & Catalog/Realtime Lead | Category-Locked Catalog & Floor Price (F-05..F-07), Socket.IO VIP Deal Chat & Negotiation (F-08..F-10) | WF-02: Category-Locked Product Compliance, MAP Floor-Price & Socket.IO VIP Deal Negotiation | Decision D2; Products, Variants, Floor Price & Chat Negotiation |
| **Nguyễn Đình Tuấn** | Fullstack Developer & Attribution Lead | 14-Day Sample SLA (F-11..F-12), Smart Link/QR & KOC Coupons (F-13), Anti-Self-Referral Guard (F-14) | WF-03: 14-Day Sample SLA, Exclusive Attribution Tools & Anti-Self-Referral Order Placement | Decision D3; Marketing Toolkit, Attribution Engine & Fraud Sentinel |
| **Nguyễn Phú Quý** | Fullstack Developer & Logistics/Arbitration Lead | Multi-Warehouse GHN Fulfillment (F-15), 14-Day Adaptive Escrow & Dual-Path Return Arbitration (F-16) | WF-04: Multi-Warehouse GHN Logistics, 14-Day Adaptive Escrow & Dual-Path Return Arbitration | Decision D4; GHN Logistics, 1-to-1 Exchange vs Refund Clawback |

# Appendix A – Defense 1.1 Targets

## A1. Main Workflows

| ID | Workflow | Actors | Trigger | Business outcome | Features | Entity states | Owner |
|---|---|---|---|---|---|---|---|
| **WF-01** | Internal Brand Staff Provisioning, KOC eKYC & Multi-Channel Brand Onboarding | System Administrator, Customer, System Manager, Collaborator | System Admin appoints an internal Brand Manager (/admin/personnel), or a Customer registers, verifies Email OTP, and submits a KOC upgrade eKYC dossier (/customer/upgrade). | In-house Shop Manager bound to assigned brand store; verified Customer upgraded to Collaborator at Bronze (10%) tier with verified social channel and immutable AuditLog. | F-01, F-02, F-03, F-04 | User: isActive=true (SHOP_MANAGER/COLLABORATOR); KycStatus: UNVERIFIED -> VERIFIED / REJECTED; Tier: Bronze (10%) | Nguyễn Thành Thắng |
| **WF-02** | Category-Locked Product Compliance, MAP Floor-Price & Socket.IO VIP Deal Negotiation | Shop Manager, System Manager, Collaborator | Shop Manager creates a 6-block product SKU, proposes a price drop, uploads Socket.IO Deal Chat & Media assets, and runs 4-Axis AI Partner Matchmaking. | Category-locked SKU moderated to APPROVED, MAP-compliant price promotion activated, Top KOC matched & approved for store cooperation, and HD media downloaded. | F-05, F-06, F-07, F-08 | ProductModerationStatus: DRAFT -> APPROVED / REJECTED; PricePromotionStatus: PENDING -> APPROVED -> ACTIVE; StoreCollaboratorStatus: PENDING -> APPROVED | Phan Xuân Thịnh |
| **WF-03** | 14-Day Sample Request SLA, Multi-Channel Attribution & Anti-Self-Referral Order Placement | Shop Manager, Collaborator, System Administrator, Customer | Approved Shop Manager and Collaborator negotiate an Exclusive VIP Deal card in Socket.IO Chat, dispatch a 0 VND product sample, and host a live product-pinning session. | VIP Deal activated (immediately for <=16% or after System Admin approval for 17%–25%), sample video verified within 14-day SLA, and live pinned SKU earns +2% live bonus. | F-09, F-10, F-11, F-12 | ExclusiveDealStatus: PENDING -> PENDING_ADMIN_APPROVAL -> APPROVED; SampleRequestStatus: PENDING -> APPROVED -> SHIPPED -> RECEIVED -> VIDEO_SUBMITTED -> COMPLETED / OVERDUE; LiveSessionStatus: SCHEDULED -> LIVE -> ENDED | Nguyễn Đình Tuấn |
| **WF-04** | Multi-Warehouse GHN Logistics, 14-Day Adaptive Escrow & Dual-Path Return Arbitration | Collaborator, Customer, Shop Manager, System Manager | Collaborator shares Smart Link/QR and KOC Attribution Coupon; Customer checks out, receives GHN parcel, and files either a damaged-item Exchange or a Refund return. | Order attributed accurately (0% if self-referral), shipped via brand GHN warehouse; 1-for-1 EXCHANGE preserves 100% KOC commission (+ GHN claim) while REFUND claws back commission. | F-13, F-14, F-15, F-16 | OrderStatus: PENDING -> SHIPPING -> DELIVERED -> COMPLETED / RETURNED; ReturnRequestStatus: REQUESTED -> SHOP_APPROVED -> INSPECTING -> EXCHANGE_SHIPPED / REFUNDED -> COMPLETED; CommissionStatus: PENDING -> APPROVED / REVERSED | Nguyễn Phú Quý |
| **WF-05** | Two-Stage Escrow Maturation (14-Day & KPI Deal), Concurrency Payouts & 10% PIT Withholding | Shop Manager, Collaborator, System Manager, System Administrator | 14-day post-delivery hold elapses on eligible orders, Shop Manager settles closed-month FR-09 milestones, and Collaborator submits a bank withdrawal request. | Mature commissions and qualified KPI bonuses unlock to availableBalance, monthly bonus & tier upgrade apply once, 10% flat PIT is withheld under DB row lock, and net payout disburses via BIDV Sandbox. | F-17, F-18, F-19 | CommissionStatus: PENDING -> APPROVED; DealBonusStatus: PENDING_KPI -> QUALIFIED / FORFEITED; PayoutStatus: PENDING -> PROCESSING -> APPROVED / REJECTED | Nguyễn Thành Thắng |

## A2. Demo Scenarios

| ID | Workflow | Type | Preconditions and data | Main steps | Expected result | Presenter |
|---|---|---|---|---|---|---|
| **DS-01a** | WF-01 | **Main** | Seeded System Admin, System Manager, and new Customer account; Sora Skincare store configured. | 1. Admin creates internal SHOP_MANAGER bound to Sora Skincare at /admin/personnel. 2. Customer registers, verifies OTP, and submits CCCD, MB Bank & TikTok channel at /customer/upgrade. 3. System Manager approves eKYC on Web/Mobile. | Staff account active on Sora Skincare; Customer upgraded to COLLABORATOR with KycStatus=VERIFIED, initial tier Bronze (10%), and AuditLog recorded. | Nguyễn Thành Thắng |

| ID | Workflow | Type | Preconditions and data | Main steps | Expected result | Presenter |
|---|---|---|---|---|---|---|
| **DS-01b** | WF-01 | **Exception** | Unauthenticated client sends role=SHOP_MANAGER to /auth/register; KOC submits blurred CCCD or >5 wrong OTPs. | 1. Call POST /auth/register with role=SHOP_MANAGER. 2. Enter wrong OTP 5 times. 3. System Manager rejects mismatched KYC bank/CCCD dossier with mandatory reason. | Public signup coerced to CUSTOMER (privilege escalation blocked); OTP invalidated after 5 failures; KycStatus=REJECTED with feedback and resubmission enabled. | Nguyễn Thành Thắng |
| **DS-02a** | WF-02 | **Main** | Sora Skincare (locked to Cosmetics), verified KOC profiles, and SKU floorPrice = 350,000 VND. | 1. Shop Manager creates Vitamin C Serum (450k VND, EAN-13); System Manager approves. 2. Shop Manager proposes 390k VND promotion (>= floorPrice); System Manager approves. 3. Run 4-Axis AI Matchmaker, approve KOC cooperation, and KOC downloads HD media. | Product APPROVED on storefront; promotion ACTIVE at 390k VND; AI returns ranked KOC radar scores; StoreCollaborator=APPROVED and HD asset downloaded. | Phan Xuân Thịnh |
| **DS-02b** | WF-02 | **Exception** | Shop Manager attempts cross-category SKU, duplicate EAN-13 barcode, or price drop below floorPrice (320k < 350k VND). | 1. Submit Electronics category from Cosmetics store. 2. Submit duplicate EAN-13 barcode. 3. Propose promotion price 320,000 VND below 350,000 VND MAP floor. | All three requests rejected with 400 Bad Request; product and price remain unchanged, protecting brand vertical isolation and unit margin. | Phan Xuân Thịnh |
| **DS-03a** | WF-03 | **Main** | Approved Sora–KOC partnership on Web/Mobile; approved Vitamin C Serum with sample quota > 0. | 1. Shop sends 16% VIP Deal card (KPI 30M VND) via Socket.IO Chat; KOC accepts (auto-APPROVED); also test 18% deal escalating to Admin approval. 2. KOC requests sample -> Shop dispatches GHN -> KOC submits review URL within 14d -> Shop marks COMPLETED. 3. KOC pins SKU in LIVE room. | 16% VIP deal active immediately (18% active after Admin approval); sample reaches COMPLETED within 14-day SLA; LIVE tray pins SKU with +2% bonus. | Nguyễn Đình Tuấn |
| **DS-03b** | WF-03 | **Exception** | VIP Deal proposed at 28% (>25% cap); KOC misses 14-day sample review deadline; pinned SKU stock hits 0. | 1. Propose 28% VIP Deal in chat. 2. Trigger overdue scan on sample past 14-day deadline and request another sample. 3. Set pinned SKU stock = 0 during LIVE session. | 28% deal blocked (400 Bad Request); sample transitions to OVERDUE and blocks new sample requests; out-of-stock SKU disables purchase CTA in live tray. | Nguyễn Đình Tuấn |
| **DS-04a** | WF-04 | **Main** | Verified KOC with active 16% VIP deal & KOC coupon (10% off); authenticated Customer; Sora GHN warehouse. | 1. KOC generates /r/:shortCode, QR & KOC coupon. 2. Customer clicks link / applies KOC coupon, logs in preserving 100% cart, pays via PayOS/COD. 3. Shop creates GHN waybill -> DELIVERED. 4. On transit damage, Customer chooses EXCHANGE -> Shop ships 1-1 replacement & opens GHN claim. | Order attributed to KOC with pending commission recorded; replacement delivered (COMPLETED), 100% KOC commission preserved, and GHN compensation claim opened. | Nguyễn Phú Quý |
| **DS-04b** | WF-04 | **Exception** | KOC attempts self-purchase via own link/coupon; Customer requests full return & refund within 14-day window. | 1. KOC checks out using own referral link/matching phone. 2. Separate Customer files REFUND return with unboxing video -> parcel returned & inspected -> refund approved. | Self-referral order sets commission rate = 0% and flags AiFraudService; refunded order credits Customer shoppingBalance and reverses KOC commission to REVERSED. | Nguyễn Phú Quý |
| **DS-05a** | WF-05 | **Main** | Delivered order past 14-day hold with no open return; closed-month clean GMV > 50M VND; verified KOC wallet. | 1. Reconciliation job matures 14-day PENDING commission to availableBalance and qualifies KPI deal escrow. 2. Shop settles closed-month FR-09 milestone bonus & promotes KOC tier. 3. KOC requests 2,072,000 VND withdrawal -> System Manager approves via BIDV Sandbox. | Wallet debited once under SELECT ... FOR UPDATE; 207,200 VND (10% PIT) posted via FinancePosting; 1,864,800 VND net approved with SBX-... reference and AuditLog. | Nguyễn Thành Thắng |
| **DS-05b** | WF-05 | **Exception** | SHOP_MANAGER attempts withdrawal; KOC requests < 200,000 VND or fires parallel requests exceeding balance; BIDV Sandbox timeout. | 1. Call POST /payouts/withdraw as SHOP_MANAGER. 2. Submit 100,000 VND (<200k min) and 2 concurrent full-balance requests. 3. Simulate BIDV timeout/rejection. | SHOP_MANAGER blocked (403 Forbidden); <200k rejected (400); row lock allows at most 1 request (0 negative balance); rejected payout restores 100% funds via PAYOUT_REJECT_REFUND. | Nguyễn Thành Thắng |

## A3. Functional Test Case Target

| Source | Count | Minimum per item | Subtotal |
|---|---|---|---|
| **Mandatory features (from Section 6)** | 19 | 3 | 57 |
| **Main workflows (end-to-end, from A1)** | 5 | 1 | 5 |
| **State transitions of main entities (KYC, Product, PricePromo, Deal, Sample, Live, Order, Return, Commission, Payout)** | 13 | 1 | 13 |
| **Agreed NFRs & concurrency/security suites (from Section 7)** | 6 | 1 | 6 |
| **Committed total (minimum)** | — | — | **81** |